

Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values entirely within normal reference ranges, with no abnormalities flagged. The predictive model assesses the profile as within normal limits.

Detailed Test Explanations

SERUM BILIRUBIN (TOTAL): Measures total bilirubin, a byproduct of red blood cell breakdown; elevated levels may indicate liver dysfunction, hemolysis, or biliary obstruction. Patient result: 0.9 mg/dl (reference: 0.2-1.2 mg/dl), normal.

SERUM BILIRUBIN (DIRECT): Quantifies conjugated bilirubin; increases suggest hepatic or obstructive issues. Patient result: 0.2 mg/dl (reference: 0-0.3 mg/dl), normal.

SERUM BILIRUBIN (INDIRECT): Reflects unconjugated bilirubin; elevations can signal hemolysis or Gilbert syndrome. Patient result: 0.70 mg/dl (reference: 0.2-1.0 mg/dl), normal.

SGPT (ALT): Alanine aminotransferase, a sensitive marker of hepatocellular injury; rises with liver cell damage from toxins, viruses, or alcohol. Patient result: 36 U/L (reference: 13-40 U/L), normal.

SGOT (AST): Aspartate aminotransferase, found in liver and other tissues; elevations indicate potential liver, cardiac, or muscle injury. Patient result: 32 U/L (reference: 0-37 U/L), normal.

SERUM ALKALINE PHOSPHATASE: Enzyme elevated in biliary obstruction, bone disease, or pregnancy; result not available.

SERUM PROTEIN: Total protein reflects nutritional status and liver synthetic function; low levels may indicate malnutrition or chronic liver disease. Patient result: 7.2 g/dl (reference: 6.4-8.3 g/dl), normal.

SERUM ALBUMIN: Primary protein synthesized by the liver; hypoalbuminemia signals chronic liver disease, nephrotic syndrome, or malnutrition. Patient result: 4.7 g/dl (reference: 3.5-5.2 g/dl), normal.

GLOBULIN: Immunoglobulins and other proteins; abnormalities may point to inflammation, infection, or liver dysfunction. Patient result: 2.50 g/dl (reference: 1.8-3.6 g/dl), normal.

A/G RATIO: Albumin-to-globulin ratio; derangements suggest chronic inflammation, liver disease, or nephrotic syndrome. Patient result: 1.88 (reference: 1.1-2.4), normal.

Overall Interpretation

All performed liver function tests, including bilirubin fractions, transaminases, proteins, and albumin-globulin ratio, fall within normal limits, consistent with preserved hepatic synthetic and metabolic function. The absence of alkaline phosphatase data does not alter this profile. The predictive model assesses the profile as within normal limits; clinical correlation is advised per laboratory remarks.

Recommendations

Maintain a balanced diet rich in fruits, vegetables, and lean proteins; engage in regular physical activity (at least 150 minutes weekly); limit alcohol intake and avoid hepatotoxic substances. Routine annual health screenings recommended. Consult a physician for any symptoms or personalized advice.

Disclaimer

This report provides interpretive guidance based on provided data and is not a substitute for professional medical evaluation, diagnosis, or treatment.